- **Never run `apt-get update` or `apt-get upgrade`.** Stack is self contained and updating software might induce compatibility issues.

- **The code is modular.** Rewrite individual services if needed, but do not attempt to rewrite the entire flash memory.

- **Do not remove the USB cable during flashing a service** (e.g., `voxl-tflite-server`, `voxl-docker-mavsdk-python`). You risk bricking the software otherwise.

- **Always put the propeller protectors back on when needed.**

- **Charging order matters.** When charging, first insert the small balance cable into the LiPo charger (on the left), then the larger power cable. When removing, disconnect the power cable first, then the small balance cable.

- **Monitor thermals with `voxl-inspect-cpu`** to check CPU and GPU load. When the drone is not propelling but services are running, it can get hot, so use the external fan to cool it.

- **Experiment startup sequence:** First take off in manual or position mode. If you started in manual, switch to position mode once the drone is stable, then switch to offboard mode before executing your scripts. Always check `voxl-inspect-qvio` to confirm QVIO quality is stable.

- **The VIO coordinate frame origin is set at the location where you power up the drone.** After powering up and before takeoff, you can walk the drone around a bit to grab more features and improve QVIO quality.

- **Never work alone.** All experiments require at least two people. One person holds the remote and is ready to manually override if necessary.

- **The kill switch instantly cuts all power, but avoid using it if the drone is too high — it will fall and be damaged. If the drone is high, switch to manual mode immediately and gently bring it back down.**

- **When making service changes, you can rely on LLMs, but verify every change.** Make sure there is no catastrophic or major modification. Understand what counts as a safe change and use LLMs with caution.

- **Inside the `voxl-mavsdk-python` Docker container, you must start the MAVSDK server with `./start_mavsdk_server.sh` first, otherwise your Python scripts will not work.**

- If you need to change propellers due to some accident, we have some spare. **Be careful with their order.** Take before removing the old ones and compare.

- **Use the ModalAI forum (https://forum.modalai.com/) and docs (https://docs.modalai.com/), or ask me (Ege Yuceel) if you run into a problem.** Use LLMs as tools for obtaining information, but be careful when editing code.

- **`/etc/modalai`** contains all the important config files for the tflite server, MAVLink server, camera server, etc.

- **`/data/modalai`** contains the camera intrinsics if ever needed.

- **The drone needs an SSID and password to connect to a network** (e.g., `ViconRouter_2.4G`). IllinoisNet will not work because it requires a username.

- **Monitor the battery** with `voxl-inspect-battery` or `voxl-portal`. To use voxl-portal, you must be on the same network as the drone — type the drone's IP address into your browser.

- **Pushing a new neural network to the drone:** if you want to deploy a newly trained, tflite-converted, float16-quantized network, connect over USB and run `adb push /directory/network.tflite /usr/bin/dnn` from your laptop (not the VOXL). Here `/usr/bin/dnn` is the target directory on the VOXL, and `/directory/network.tflite` is the network on your laptop.

- **Recording camera images:** SSH into the drone and run `voxl-record-raw-image hires_small_color -d /data/raw_capture/ -n 300` (records at 30–40 Hz). Then archive with `tar -cvf raw_capture.tar raw_capture/` and pull to your laptop with `adb pull /data/raw_capture.tar`. Use the `bintojpg.py` script to convert the raw files to images.